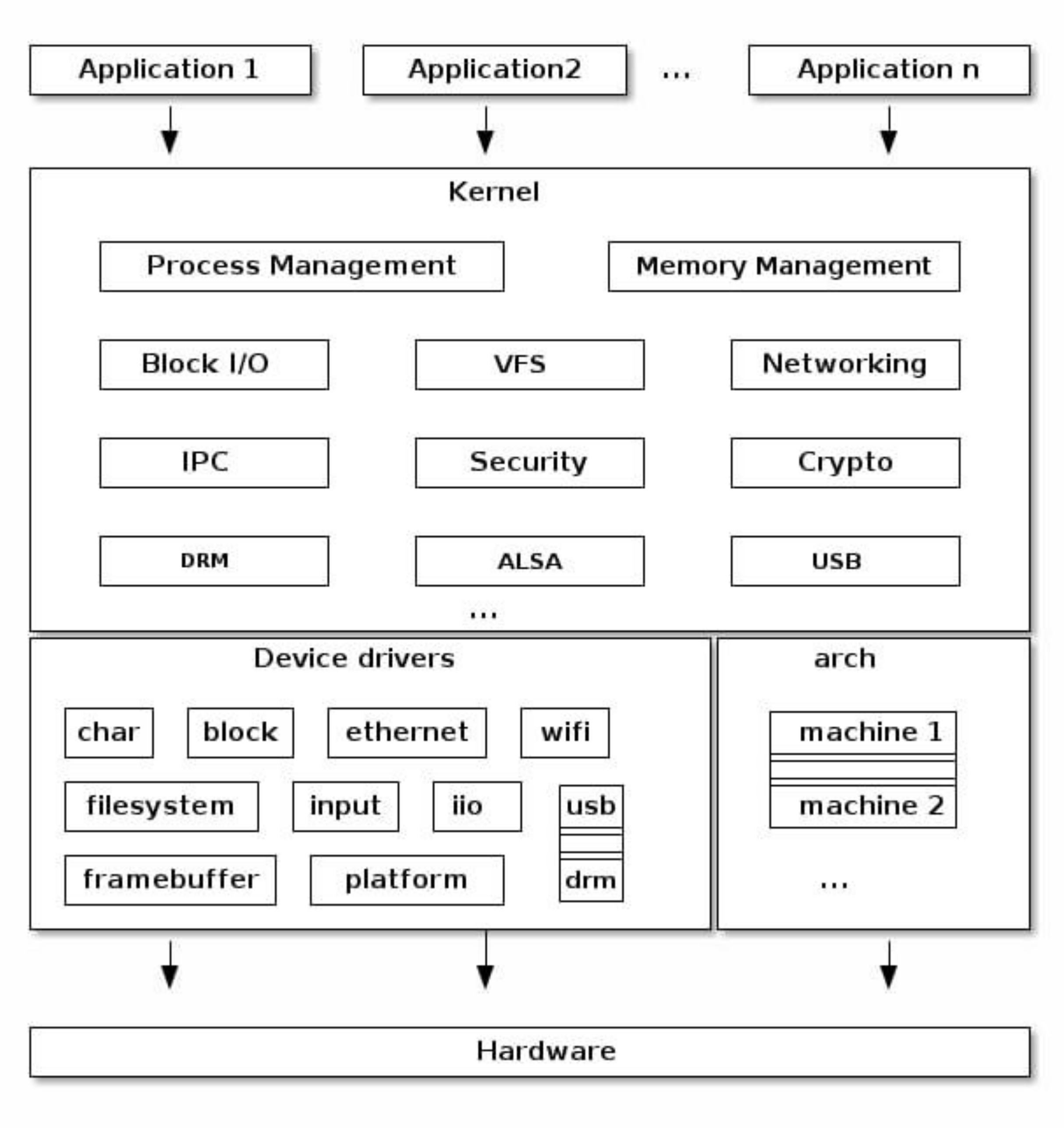
# NAME - Shipra Pal
# REGISTRATION NO.-2141016019

# TASK 1

**Kernel Architecture Diagram - Draw a detailed diagram of the Linux kernel architecture. Label and write a short description (2-3 sentences) for each major component like Scheduler, File System, Network Stack, etc.**



## Component Description :

1. System Call Interface (SCI) -

→ Acts as a gateway between user space and kernel space.

→ It translates user-level function calls into kernel-level operations.

2. Process Scheduler :

→ Manages process execution and CPU time allocation.

→ It decides which process runs when and for how long, using various scheduling algorithms.

3. Memory Manager -

→ Handles physical and virtual memory allocation.

→ Includes feature like paging, swapping, and memory mapping to provide efficient memory usage.

4. File System -

→ Manages data storage and retrieval.

→ Provides a structured way to access files and directories, supporting multiple file system like ext4, XFS, and more.

5. Network Stack -

→ Handles all networking functionalities.

→ Includes support for protocols like TCP/IP, socket communications, and network device interfacing.

6. Device Drivers and Modules -

→ Serve as intermediaries between the kernel and hardware devices.

→ Drivers allow the kernel to communicate with hardware like disk drivers, GPUs and peripherals

7. Architecture - Dependent Layer -

→ Contains code specific to hardware. Architecture.

→ It abstracts the hardware differences from the rest of the kernel.